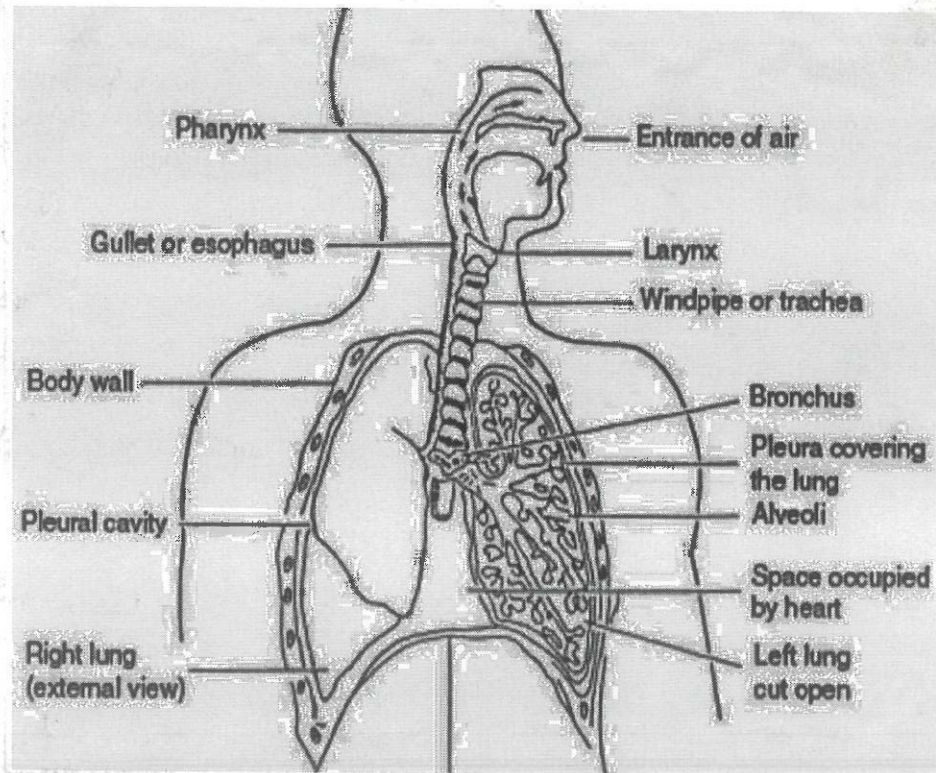


Measurements in Respiratory System

3.6 Physiology of Respiratory System



- Respiration is the exchange of gases in any biological process.
- The Nasal Cavities, Larynx, Trachea, Bronchi and Bronchioles are the body parts through which air enters the lungs.
- The lungs are like elastic bags located in a closed cavity called Thoracic Cavity.
- The right lung consists of three lobes i.e; upper, middle and lower lobes, and Left lung has two lobes upper & lower.
- The Larynx, known as the Voice box, is connected to bronchi through trachea; Sometimes called the wind pipe.
- Whenever a person swallows the food and liquids are directed to the esophagus that is connected to Stomach.
- The trachea is about 1.5 to 2.5 cm in diameter and approximately 11 cm long, extending from the larynx to the upper boundary of chest.

→ The Pleura a thin membrane covers the lungs, which passes from the lungs at its root into the interior of the chest wall and upper surface of the diaphragm.

→ The diaphragm is located at the bottom of the thoracic cavity, which when pulled downward to enlarge the thorax.

→ This action is the principal force involved in inspiration or inhaling.

→ The interchange of the oxygen from the lungs to the blood, the diffusion of CO_2 from the blood to the lungs takes place in the capillary surfaces of the Alveoli.

3.7 Tests and Instrumentation for the Mechanics of Breathing:-

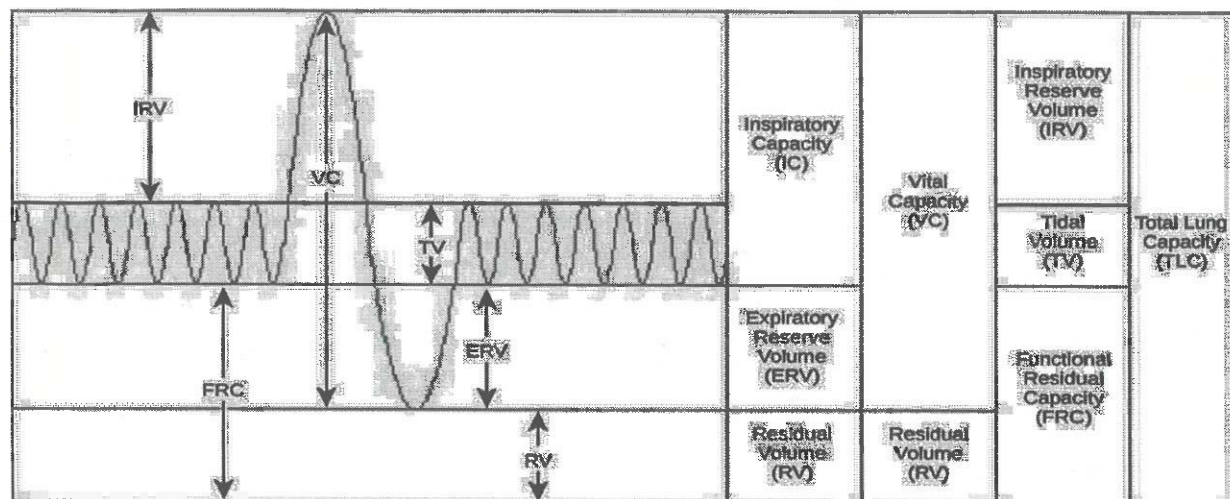
The mechanics of breathing concern the ability of a person to bring air into his lungs from the outside atmosphere and to exhaust air from the lungs.

This ability is affected by the various components of the air passages, the diaphragm and associated muscles, the rib cage and associated musculature, and the characteristics of the lungs themselves.

There are number of most prominent measurements and tests that are used clinically and in research in connection with the mechanics of breathing.

Sometimes the instrumentation is also required for these tests and measurements.

3.7.1 Lung Volumes and Capacities :-



Tidal Volume [TV] :- It is the volume of gas inspired or expired during each normal respiration cycle.

Inspiratory Reserved Volume [IRV] :- It is the extra volume of gas that a person can inspire with maximum effort for reaching normal inspiratory level.

Expiratory Reserved Volume [ERV] :- It is the extra volume of gas that can be expire with maximum effort beyond the expiratory level.

Residual Volume [RV] :- It is the volume of gas remaining in the lungs at the end of maximal expiration.

Virtual Capacity [VC] :- It is the maximum volume of gas that can be expell from the lungs by force full effect after a maximal inspiration. It is also the sum of TV, IRV and RV.

Total Lung Capacity [TLC] :- It is the amount of gas contained in the lung at the end of maximal inspiration. It is the sum of VC and RV. It is also the sum of TV, IRV, ERV and RV.

Inspiratory Capacity [IC] :- It is the maximum amount of gas that can be inspired after reaching the end expiratory level. It is also sum of TV and IRV.

Functional Residual Capacity [FRC] :- It is the volume of gas remaining in the lungs at the end of expiratory level. It is also the sum of RV and ERV. It can also be calculated as Total Lung Capacity minus Inspiratory Capacity.

Forced Vital Capacity [FVC] :- It is the total amount of air that can be forcibly expired as quickly as possible after taking the deepest possible breath.

Forced Expiratory Volume [FEV] :- Maximum amount of gas that can be expelled in a given no. of seconds.

Maximal Expiration flow [MEF] :- It is the rate during the first litre expired after 200ml has been exhausted at the beginning of the FEV.

Maximal Breathing Capacity [MBC] :- It is the measure of maximum amount of air that can be breathed in and blown out at certain interval such as 15 (or) 20 Sec.

Most of the air enters the lungs to fill the alveoli after each breath.
→ A certain amount of air is required to fill the various cavities of the air passages. This air is called the Dead-Space air, and the space it occupies is called the Dead-Space.

3.7.2 Mechanical Measurement :-

- Determination of Compliance, is defined as the Volume increase in the lungs per unit increase in lung pressure.
- In practice, two types of Compliance measurements are there. They are Static and Dynamic.
- Static Compliance is determined by obtaining a ratio of the difference in lung Volume at two different Volume levels and the associated difference in lung alveolar pressure.
- Dynamic Compliance is measured by the Tidal Volume.
- Lung Compliance varies with the Size of lungs; a child has a Smaller Compliance than an adult.
- Compliance does not remain Constant over the breathing cycle, but tends to decrease as the lungs are inflated.

3.7.3 Instrumentation for Measuring the Mechanics of Breathing :-

The most widely used laboratory instrument for respiratory Volume measurements is the recording Spirometer. All the lung Volumes and Capacities that can be determined by measuring the amount of gas inspired or expired under a given set of Conditions can be obtained by use of the Spirometer.

- The Standard Spirometer consists of a movable bell inverted over a chamber of water.
- Inside the bell, above the water line, is the gas that is to be breathed.
- The bell is balanced by a weight to maintain the gas inside at atmospheric pressure so that its height above the

Water is proportional to the amount of gas in the bell.

→ A breathing tube connects the mouth of the patient with the gas under the bell.

→ In this case nose of the patient is blocked.

→ As the patient breaths into the tube, the bell moves up and down with each inspiration and expiration in proportion to the amount of air breathed in or out.

→ A pen attached to the balancing weight mechanism writes on the paper attached to the drum recorder called a "kymograph".

→ As the kymograph rotates, the pen traces the breathing pattern of the patient.

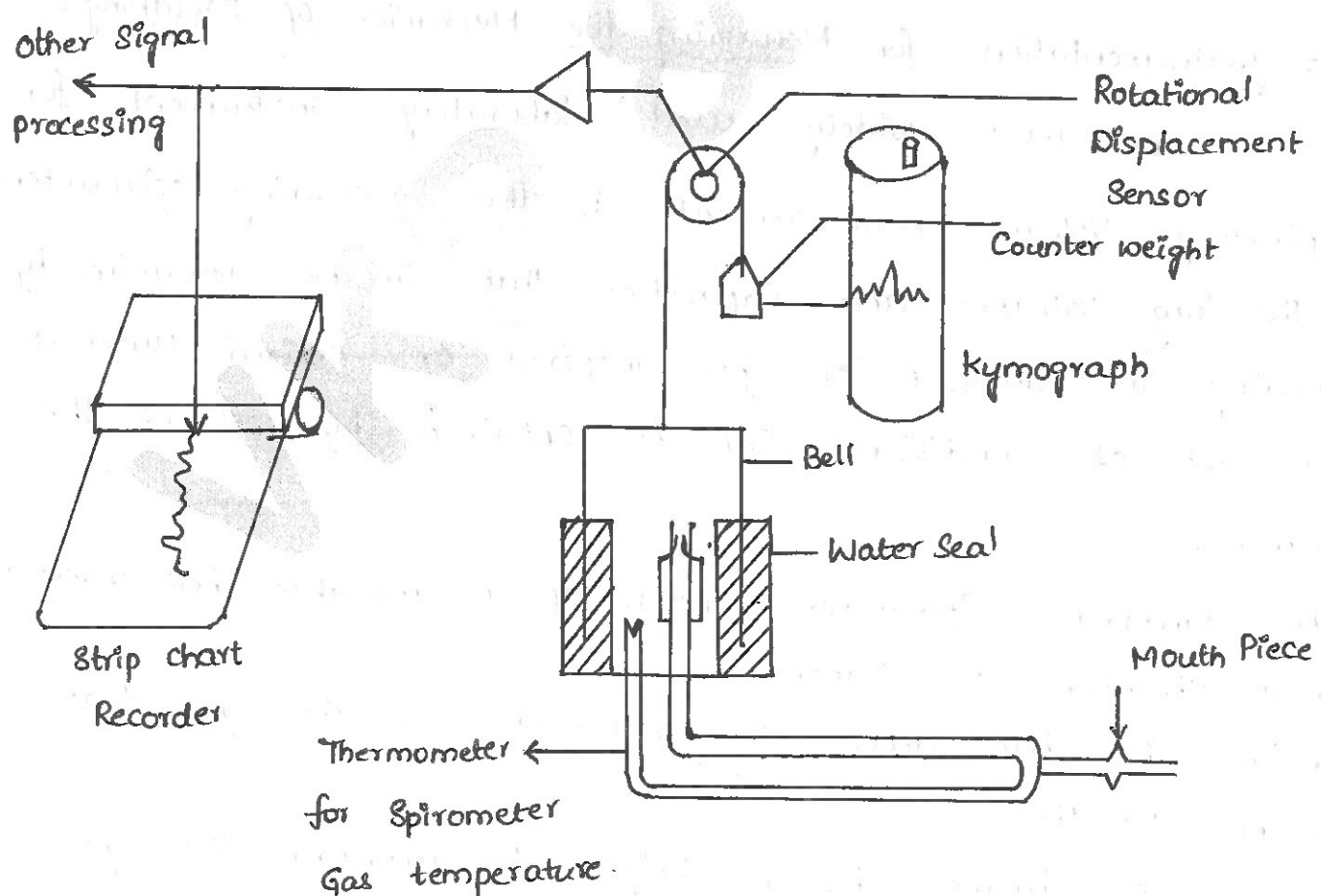
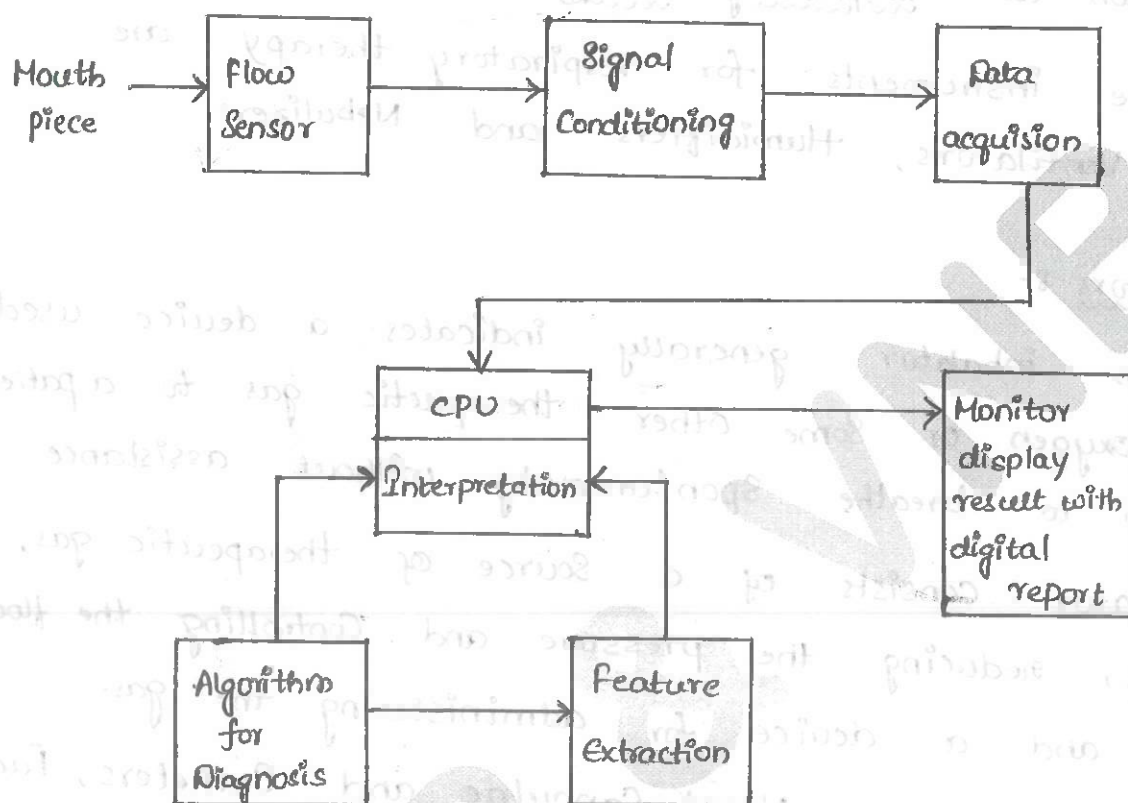


fig:- Water Sealed Spirometer

Electronic Spirometer :-

→ These electronic Spirometers measures airflow by the use of electronic circuitry calculates the various Volumes and Capacities.



→ This instrument provides both a graphic output similar to that of a Standard Spirometer.

→ It also gives the digital read out of the desired parameters.

→ Some instruments even have built in Computational Capabilities to calculate automatically the required Volumes and Capacities from the basic measurements.

→ These Spirometers can be used to measure the Residual Volumes of lungs and also used to determine the "Intraalveolar and Intra thoracic Pressure".

3.8 Respiratory Therapy Equipment :-

The procedure and the instrumentation required in providing mechanical assistance and higher concentration of oxygen or other therapeutic gases or medicine constitutes a field known as "Respiratory Therapy".

The instruments for respiratory therapy are Inhalators, Ventilators, Humidifiers and Nebulizers.

3.8.1 Inhalators :-

→ The term inhalator generally indicates a device used to supply oxygen or some other therapeutic gas to a patient who is able to breathe spontaneously without assistance.

→ The inhalator consists of a source of therapeutic gas, equipment for reducing the pressure and controlling the flow of the gas, and a device for administering the gas.

→ This device includes Nasal Cannulae and Catheters, Face masks that cover the nose and mouth.

→ The oxygen concentration presented to the patient is controlled by adjusting the flow of gas into the mask.

3.8.2 Ventilators & Respirators :-

→ Ventilators & Respirators are used interchangeably to describe the equipment that may be used continuously or irregularly to improve ventilation of the lungs.

→ Most of the ventilators in the clinical settings use positive pressure during inhalation to inflate the lungs with various gases or mixture of gases i.e. air, oxygen, Carbon dioxide etc;

- The Respirator helps the patient to inhale when he wants to breath.
- Controlled Ventilation is required for the patient who are unable to breath on their own.
- If the patient fails to breath with in predetermined time, a timer automatically triggers the device.
- Patient Controls his own breathing as long as he can but; if he fails to do so the machine is able to take control over him.
- This mode is most frequently used in critical care conditions.
- Ventilators in the hospitals can be divided into two types.
 - ↳ Pressure cyclic
 - ↳ Volume cyclic.

Volume cyclic respirator provides much degree of control over ventilation than Pressure cycle type.

3.8.3 Humidifiers & Nebulizers :-

- To prevent the damage to the patient lung; the air (or) oxygen given during respiratory therapy must be humidified.
- Virtually all Inhalators, Ventilators and Respirators must include equipment to humidity the air either by heat Vapourization (or) by bubbling an air stream through a jet of water.
- If the therapy requires the water (or) some type of medication; a device called a Nebulizer is used.
- Ultrasonic nebulizers are more effective and more expensive type of nebulizers.

Important Questions

1. Draw the Einthoven triangle? [April/May-2019] [Set-1]

Ans: Refer Section 3.1.3.

2. What are the different ECG lead Configurations? Explain [April/May-2019] [Set-1], [Nov-2020]

Ans: Refer Section 3.1.3.

3. Draw the Schematic of ECG recorder by showing all building blocks. [April/May-2019] [Set-1]

Ans: Refer Section 3.1.4.

4. Explain the operation of Spirometer? [April/May-2019]

Ans: Refer Section 3.7.3

5. Draw the typical ECG signal and label it? [Apr/May-2019]

Ans: Refer Section 3.1

6. What is Vector Cardiograph? Explain [Apr/May-19] [Set-2]

Ans: Refer Section 3.1.5.

7. What is the necessity of Ventilators? Explain [Apr/May-19] [S-2]

Ans: Refer Section 3.8.2

8. Explain how blood pressure is measured? [Apr/May-19] [S-3]

Ans: Refer Section 3.2

9. What are different respiratory therapy equipment? Explain [Apr/May-19] [S-3]

Ans: Refer Section 3.8 full.

10. Explain the working of blood flow meter? [Apr/May-19] [S-3]

Ans: Refer Section 3.3 [Any one Method].

11. Explain various tests and instrumentation for breathing? [Nov-2018]

Ans: Refer Section 3.7 full.

12. Explain about ECG lead Configurations. [NOV-2020]

Ans: Refer Section 3.1.3.

13. Discuss various methods of direct measurements of arterial blood pressure? [NOV-2020]

Ans: Refer Section 3.2.3.

14. Explain the measurement process of lung function.
[NOV-18]

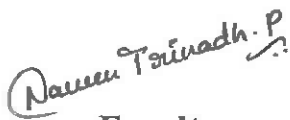
Ans: Refer Section 3.7.1.

15. Explain various tests & instrumentation for breathing?
[NOV-18]

Ans: Refer Section 3.7.3

16. Explain clearly the method of heart sound measurement?
[NOV-17]

Ans: Refer Section 3.4.


Naveen Tevinadh.P.

Faculty



HOD

Director

Principal

